MA378 Chapter 5: Review

§5.1 Review (and Exam Preview)

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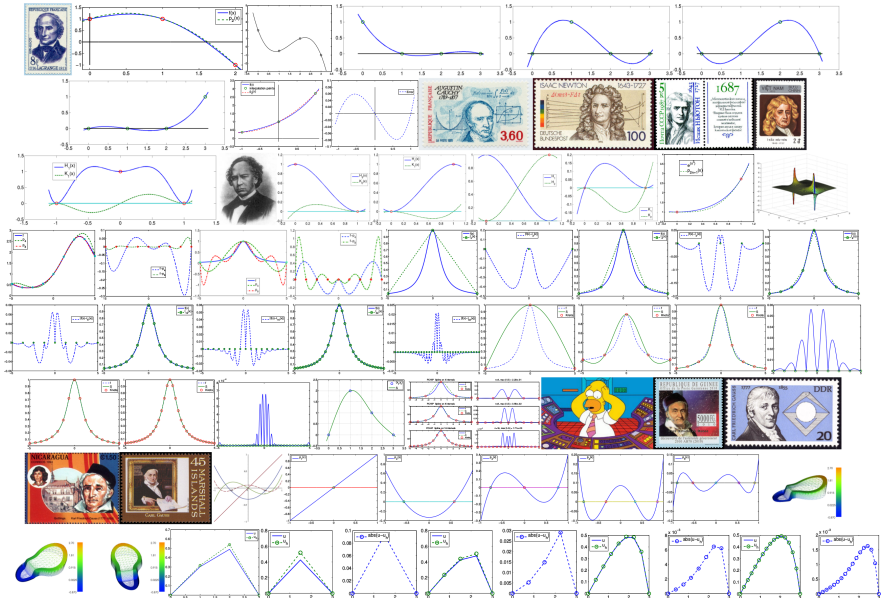

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1.1 Review

Check List

This module has four chapters:

1. Polynomial Interpolation
2. Piecewise polynomial interpolation
3. Numerical Integration (Quadrature)
4. Finite element methods

We also studied implementation of methods in MATLAB.

We now review these — primarily to provide you with a check-list of topics.

1.1 The polynomial interpolation problem (PIP) in 2 forms;

1.2 Uniqueness (*if* $p_n \in \mathcal{P}_n$ *has* $n+1$ *zeros, then* $p_n \equiv 0$)

1.2 **Lagrange Polynomials and Lagrange Interpolation**

1.3 Rolle's Theorem

1.3 **Cauchy's theorem**: $f(x) - p_n(x) = \frac{f^{(n+1)}(\tau)}{(n+1)!}\pi_{n+1}(x)$.

1.4 Hermite Interpolation: construction and error estimates.

1.5 Convergence & Runge's Example

2.1 Linear Interpolating Splines:

- construction,
- error analysis,
- best approximation, and
- the Minimum Energy property.

2.2 (Natural) Cubic Splines: construction (error analysis stated without proof)

2.3 PCHIP: Piecewise Cubic Hermite Interpolating Polynomial.

- construction,
- error analysis.

3.1 Newton-Cotes methods, and the Trapezium Rule

3.1 Undetermined Coefficients

3.2 Simpson's Rule and (non-sharp) error estimates

3.3 Precision

3.3 Composite rules

3.4 Gaussian Quadrature (and *Undetermined Coefficients* again)

3.5 A sequence of Orthogonal Monic Polynomials (and vector spaces and inner products); properties of orthogonal monic polynomials; their roots as quadrature points.

3.6 Precision and convergence of Gaussian Quadrature methods.

1.2 §4: Finite Element Methods

4.1 Boundary value problems, and Maximum Principles.

4.1 The **variational** (weak) **formulation** of the problem, and the uniqueness of its solution.

4.2 The **FEM**, including finite and infinite dimensional spaces.

4.2 **FE implementation**, including Galerkin basis functions ("hat functions"), how to construct the linear system, and why the system matrix is tridiagonal and symmetric.

4.3 Analysis, **Cea's Lemma**.

1.2 §4: Finite Element Methods

Context

Numerical Analysis is an ancient field of mathematics. The origins of the topics we've studied have roots in astronomy: interpolation techniques were used in astronomy. Earliest known examples date from use of linear interpolation in 200 BC (Babylon; modern Iraq), polynomial interpolation in 600 AD (China), and 650 AD (India).

Quadrature (numerical integration) has been studied for about as long – and long before the invention/discovery of calculus. The Trapezium Rule was known to the Babylonians, and was used by the ancient Greeks for tasks such as estimating π.

Finite Elements have a more recent history, and have been developed since the 1950, starting with the work of Richard Courant.

1.2 §4: Finite Element Methods

All three area feature in modern mathematical research.

- Since about 2010 there has been reviewed interest in robust methods for very high-order polynomial interpolation;
- Modern research in quadrature focuses on fast methods for high-dimensional problems;
- 1,000s of papers are published every year on finite element methods, with much effort aimed at understanding how to construct meshes automatically, use of discontinuous basis functions, and applications to non-linear problems.

So: in spite of the long history, what you have studied is very relevant to modern mathematics.

FYI: The monic polynomials we studied in Chapter 3 are also known as Legendre Polynomials, named after Adrien-Marie Legendre.
This is the only known "portrait" of him, which was (re)discovered in 2008.

1.3 Assessment

MA378 is assessed by

- Labs (10%)
- Two Assignments (20%)
- Class test (10%)
- Written two-hour exam (60%)

The exam will have **4 questions**: answer all 4. Q1 (28 marks) has 4 parts, all concerning theorems and proofs. **Answer any two**. For the Semester 2 2022/2023 exam, the theorems in question are selected from the following:

- **Chapter 1**:
  - 2.3+2.7 (There exists a unique solution to the PIP);
  - 3.3 (Cauchy's Theorem).
- **Chapter 2**:
  - 1.7 (Error is no worse than twice the best possible);
  - 1.9 (Minimum Energy)
- **Chapter 3**:
  - 5.8+5.10 (Monics form a basis, and are orthogonal to all polys of lower degree),
  - 5.15 (Zeros of the monic polys have special properties).
- **Chapter 4**:
  - 1.2+1.3 (Max Principle, and unique solution to the BVP),
  - 1.6 (Unique solution to the variational problem).

The remaining 3 questions all are worth 24 marks.

Q2: Interpolation (all types)

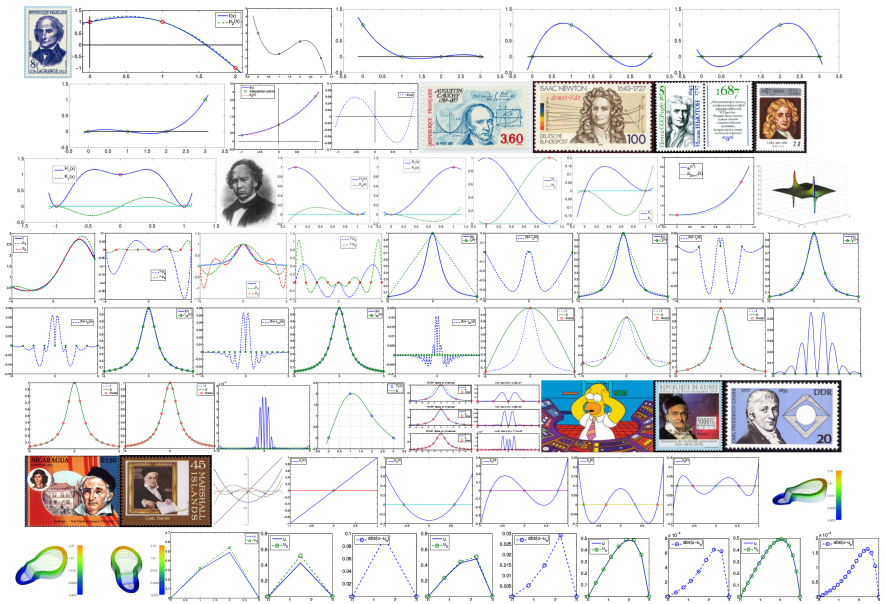
Q3: Quadrature (all types)

Q4: BVPs+FEMs

For each of these, the emphasis is on derivation of methods, and demonstration of understanding, rather than proving theorems (except the occasional uniqueness proof...).

The marks for each sub-section is indicated on the paper. Pay attention to those, since they indicate, roughly, the effort involved.

1.4 Thanks

Thanks for taking this module. I hope you have found it interesting, and that some of the topics covered are useful in your future mathematical and computational efforts.

Feedback from graduates suggests its one of the most useful "MA" modules. If you are interested in graduate studies in this field, in Ireland or elsewhere, please talk to me.